Math 190-1: Mathematics of Sustainability **Spring 2026**

Lecture 19: Limit Distributions

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19.1 Limit Distributions

Example 19.1 (Ehrenfest chain). *This chain originated in physics as a model for two cubical volumes of air connected by a small hole. In the mathematical version, we have two "urns," i.e., two of the exalted trash cans of probability theory, in which there are a total of N balls. We pick one of the N balls at random and move it to the other urn. Let X_n be the number of balls in the "left" urn after the n^{th} draw. It should be clear that X_n has the Markov property; i.e., if we want to guess the state at time $n+1$, then the current number of balls in the left urn X_n, is the only relevant information from the observed sequence of states $X_n, X_{n-1}, ...X_1, X_0$. To check this we note that:*

$$\mathbb{P}(X_{n+1} = i+1 | X_n = i, X_{n-1} = i_{n-1}, \cdots, X_0 = i_0) = \frac{N-i}{N}$$

since to increase the number we have to pick one of the $N-i$ balls in the other urn. The number can also decrease by 1 with probability i/N . In symbols, we have computed that the transition probability is given by

$$p(i, i+1) = \frac{N-i}{N}, \quad p(i, i-1) = \frac{i}{N} \quad for 0 \leq i \leq N.$$

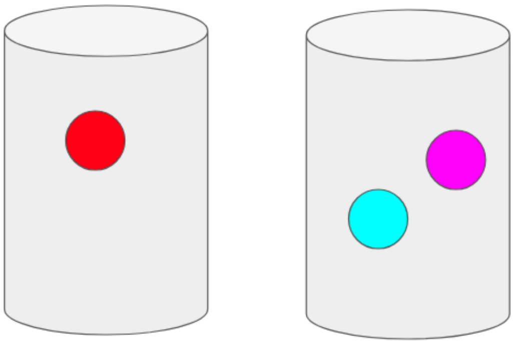
with $p(i,j) = 0$ otherwise. When $N = 3$, we get

	0	**1**	**2**	**3**
0	0	3/3	0	0
1	1/3	0	2/3	0
2	0	2/3	0	1/3
3	0	0	3/3	0

In the second power of p the zero pattern is shifted:

	0	**1**	**2**	**3**
0	1/3	0	2/3	0
1	0	7/9	0	2/9
2	2/9	0	7/9	0
3	0	2/3	0	1/3

To see that the zeros will persist, note that if we have an odd number of balls in the left urn, then no matter whether we add or subtract one the result will be an even number. Likewise, if the number is even, then it will be odd on the next one step. This alternation between even and odd means that it is impossible to be back where we started after an odd number of steps. In symbols, if n is odd then $p^n(x,x) = 0$ for all x.



	0	1	2	3
0	0	3/3	0	0
1	1/3	0	2/3	0
2	0	2/3	0	1/3
3	0	0	3/3	0

Figure 19.1: Let's consider this the set-up to the problem, i.e. $X_0 = 1$ since there is one ball in the left urn.

When $N = 3$, we get

$$\begin{array}{ccccc} & \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{0} & 0 & 3/3 & 0 & 0 \\ \mathbf{1} & 1/3 & 0 & 2/3 & 0 \\ \mathbf{2} & 0 & 2/3 & 0 & 1/3 \\ \mathbf{3} & 0 & 0 & 3/3 & 0 \end{array}$$

In the second power of p the zero pattern is shifted:

$$\begin{array}{ccccc} & \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{0} & 1/3 & 0 & 2/3 & 0 \\ \mathbf{1} & 0 & 7/9 & 0 & 2/9 \\ \mathbf{2} & 2/9 & 0 & 7/9 & 0 \\ \mathbf{3} & 0 & 2/3 & 0 & 1/3 \end{array}$$

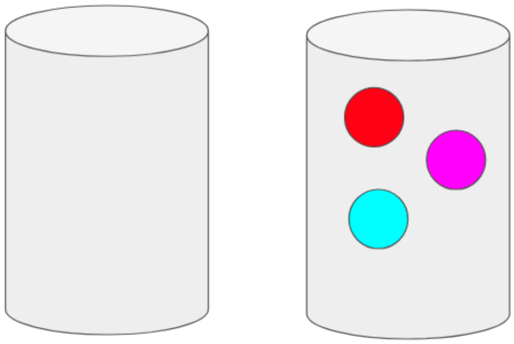
Question 1. *Will this persist?*

- This alternation between even and odd means that it is impossible to be back where we started after an odd number of steps. You need an even number of steps to get back.
- In the second power of p the zero pattern is shifted:

$$p^2 = \begin{array}{ccccc} & \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{0} & 1/3 & 0 & 2/3 & 0 \\ \mathbf{1} & 0 & 7/9 & 0 & 2/9 \\ \mathbf{2} & 2/9 & 0 & 7/9 & 0 \\ \mathbf{3} & 0 & 2/3 & 0 & 1/3 \end{array}$$

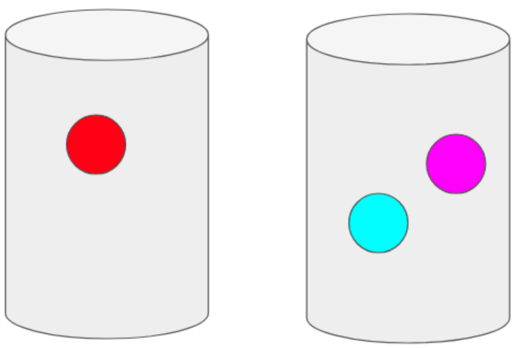
- In symbols, if n is odd then $p^n(x,x) = 0$ for all x.

Definition 19.2. *The period of a state is the largest number that will divide all the $n \geq 1$ for which $p^n(x,x) > 0$. That is, it is the greatest common divisor of $I_x = \{n \geq 1 : p^n(x,x) > 0\}$.*



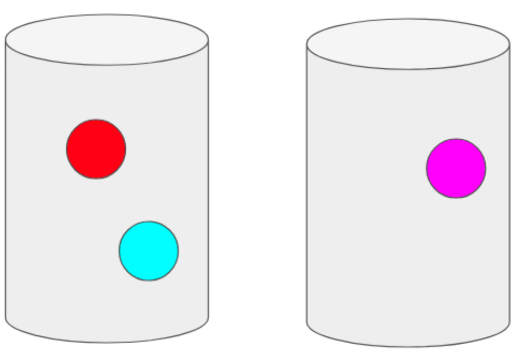
	0	1	2	3
0	0	3/3	0	0
1	1/3	0	2/3	0
2	0	2/3	0	1/3
3	0	0	3/3	0

Figure 19.2: $\frac{1}{3}$ of the time, the only ball in the left urn will be selected and moved to the right urn.



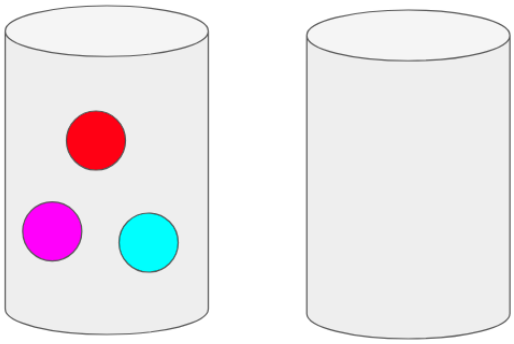
	0	1	2	3
0	0	3/3	0	0
1	1/3	0	2/3	0
2	0	2/3	0	1/3
3	0	0	3/3	0

Figure 19.3: With probability 1, we return back to the initial set up after two steps.



	0	1	2	3
0	0	3/3	0	0
1	1/3	0	2/3	0
2	0	2/3	0	1/3
3	0	0	3/3	0

Figure 19.4: $\frac{2}{3}$ of the time, one ball in the right urn will be selected and moved to the left urn.



	0	**1**	**2**	**3**
0	0	3/3	0	0
1	1/3	0	2/3	0
2	0	2/3	0	1/3
3	0	0	3/3	0

Figure 19.5: $\frac{1}{3}$ of the time, one ball in the right urn will be selected and moved to the left urn.

	-2	**-1**	**0**	**1**	**2**	**3**
-2	0	0	1	0	0	0
-1	1	0	0	0	0	0
0	0	0.5	0	0.5	0	0
1	0	0	0	0	1	0
2	0	0	0	0	0	1
3	0	0	1	0	0	0

Remark 19.3. *To check that this definition works correctly, we note that in the Ehrenfest exmaple,*

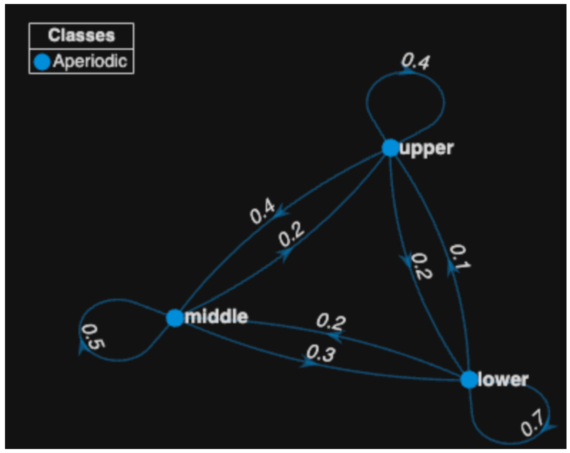
$$I_x = \{n \geq 1 : p^n(x, x) > 0\} = \{2, 4, ...\}$$

, so the greatest common divisor is 2.

Example 19.4 (Triangle and Square)**.**

- *from zero, equally likely to gor to 1 or -1.*
- *If we go -1:*
  - *we go to -2, then*
  - *we go back to 0.*
- *If we go 1:*
  - *we go to 2, then*
  - *we go to 3, then*
  - *we go back to 0.*
- *The name refers to the fact that* $0 \to -1 \to -2 \to 0$ *is a triangle and* $0 \to 1 \to 2 \to 3 \to 0$ *is a square.*

$$\begin{tikzcd}
\mathbf{-1} \arrow[d] & \mathbf{0} \arrow[l, "1/2"'] \arrow[r, "1/2"] & \mathbf{1} \arrow[d] \\
\mathbf{-2} \arrow[ur] & \mathbf{3} \arrow[u] & \mathbf{2} \arrow[l]
\end{tikzcd}$$



Question 2. *What are the possible periods of this problem? What is I_0?*

Clearly, $p^3(0,0) > 0$ and $p^4(0,0) > 0$ so $3, 4 \in I_0$. We need to think to realized that

$$I_0 = \{3, 4, 5, 6, 7, 8, 9, 10, 11, ...\}$$

While periodicity is a theoretical possibility, it rarely manifests itself in applications, except occasionally as an odd-even parity problem, e.g., the Ehrenfest chain.

Definition 19.5 (Aperiodic)**.** *In most cases we will find (or design) our chain to be* aperiodic*, i.e., all states have period 1.*

Lemma 19.6. *If $p(x,x) > 0$, then x has period 1.*

Proof: Remember the period of a state is the largest number that will divide all the $n \geq 1$ for which $p^n(x,x) > 0$. If it works for $n = 1$, then it works for all $n \geq 1$, so it have to have period 1. ■

Lemma 19.7. *If $\rho_{xy} > 0$ and $\rho_{yx} > 0$ then x and y have the same period.*

Remark 19.8. *The short answer is that if the two states have different periods, then by going from x to y, from y to y in the various possible ways, and then from y to x, we will get a contradiction.*

Theorem 19.9 (Convergence Theorem)**.** *Let p be irreducible, aperiodic, and there is a stationary distribution π. Then as $n \to \infty$, $p^n(x,y) \to \pi(y)$.*

	Lower	**Middle**	**Upper**
Lower	0.7	0.2	0.1
Middle	0.3	0.5	0.2
Upper	0.2	0.4	0.4

Example 19.10.

Question 3. *Is it irreducible?*

Question 4. *Is it aperiodic?*

19.2 Double Stochastic Chains

Definition 19.11 (Doubly Stochastic)**.** *A transition matrix p is said to be* doubly stochastic *if its COLUMNS sum to 1, or in symbols*

$$\sum_x p(x, y) = 1$$

.

Remark 19.12. *Remember this is different from our Markov matrices ("singlely stochastic") which only have their rows sum to 1, $\sum_y p(x, y) = 1$.*

Theorem 19.13. *If p is a doubly stochastic transition probability for a Markov chain with N states, then the uniform distribution, $\pi(x) = 1/N$ for all x, is a stationary distribution.*

Proof: To figure this out, we need to remember that $\pi p = \pi$ is the formula for a stationary distribution. If we assume $\pi(x) = 1/N$, then

$$\pi p = \sum_x \pi(x)p(x, y) = \frac{1}{N}\sum_x p(x, y) = \frac{1}{N} = \pi(y).$$

■

Example 19.14 (Symmetric random walk (with bumpers))**.** *Let's say the states are $\{0, 1, 2, ..., L\}$. The chain goes to the right or left at each step with probability $1/2$. However, if you are at 0 or L, and try to go out of bounds, the bumpers keep you at the same state.*

	0	**1**	**2**	**2**	**2**
0	0.5	0.5	0	0	0
1	0.5	0	0.5	0	0
2	0	0.5	0	0.5	0
3	0	0	0.5	0	0.5
4	0	0	0	0.5	0.5

Question 5. *What is the stationary distribution?*

Consider a circular board game with only six spaces {0, 1, 2, 3, 4, 5}. On each turn we roll a die with 1 on three sides, 2 on two sides, and 3 on one side to decide how far to move.


Tiny Board Game
0
1
2
3
4
5